Ameriforge

An AFG Holdings Company

Material Test Report

Heat Code: ET6

ISO 9001:2015 Certified

Ameriforge Ltd
13770 Industrial Rd. Houston, TX 77015
Sales: (713) 393-4200

NOV-R&M ENERGY SYS 15278	PO: 4032305	Sales Order: 189994	Line: 1
500 CONROE PARK WEST DR	Item Code: 9999H61002-0031R	Qty Shipped: 951	
CONROE, TX 77303	Item Desc: 2.00 150-900 HUB A350LF2 -17518204-001 NPM CDFRG		
	Mill Heat: C217479		

Spec: ASME SA350/SA350M(21) LF2 CL1 & NOV 61-0072 Rev 01 (Hubs)

Element	(%wt)	Ladle	Product	EPCRA	CAS#
C	Carbon	0.19	0.17		
Mn	Manganese	1.15	1.21	Y	7439-96-5
P	Phosphorous	0.008	0.008		
S	Sulphur	0.01	0.01		
Si	Silicon	0.27	0.28		
Cu	Copper	0.15	0.15	Y	7440-50-8
Ni	Nickel	0.06	0.06	Y	7440-02-0

Element	(%wt)	Ladle	Product	EPCRA	CAS#
Cr	Chromium	0.11	0.16	Y	7440-47-3
Mo	Molybdenum	0.02	0.02		
V	Vanadium	0.04	0.03		
Cb	Columbium	0.00	0.01		
Al	Aluminum	0.02	0.02		
C.E.		0.43			

Mechanical Testing

Test Lab	AFG
Test Bar Size	SACRIFICIAL PIECE
HBW	143 - 143
Elg (%)	32.9
RA (%)	70
Tensile Specimen Size (in)	.250
Tensile (ksi)	78.7
Yield (ksi)	52.8
Gauge Length (in)	1
Specimen Orientation	LONGITUDINAL
LCVN Charpy Temp. (F)	-50
LCVN (ft lbs)	90 - 71 - 103
LAvg 3(ft-lbs)	88
L Min (ft-lbs)	71
% Shear	40 - 30 - 40
Lat. Exp.	.073 - .054 - .080

Heat Treat

Norm. Temp. (F)	1675
Norm. (Hrs)	2.30
Norm. Cooling Type	AIR

Other

Supplier	STEEL DYNAMICS
Country of Melt	USA
Country of Forging	USA
Vacuum Degassed	Y
Fully Killed	YES
Made to Fine Grain Practice	YES

Authorized Signature
By: [signature]
Date: 1/12/23

The recording of false, fictitious, or fraudulent statement or entries on this document may be punishable as a felony under federal statute.

EPCRA Supplier Notification: This product may contain one or more toxic chemicals subject to the reporting requirements of Section 313 of the Emergency Planning and Community Right-to-Know Act (Title III of the Superfund Amendments and Reauthorization Act of 1986) and 40 C.F.R. Part 372. Potentially reportable chemicals are indicated with a checkmark in the "EPCRA" column and a Chemical Abstract Services (CAS) registry number is provided for each such chemical in addition to the percent by weight of the chemical present in this product. It is your responsibility alone to determine whether your facility is required to submit a Toxic Release Inventory Report under EPCRA Section 313.

Certification No.: 1620642
Certification Date: 1/9/2023
Issued By: Michelle Gutierrez

This report is issued in compliance with the requirements of EN10204 3.1 / ISO 10474 3.1.b

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Product compliant with NACE MR0175/ISO 15156-2:2015, Annex A (A.2.1.2, A2.1.3).
Product compliant with NACE MR0103/ISO 17945:2015, Section 13.1.1 and 13.1.3.

* No Weld repair performed.
* Ladle Chemical Analysis results are reported from the raw material suppliers MTR.
* Tensile taken from a round specimen. Measurement of elongation after fracture.
* Yield strength was determined using the .2% offset method, unless specified.
* All material supplied under this order is certified to be free of mercury contamination and no mercury bearing equipment was used in manufacturing, fabrication or testing.
* When reported on MTR: Tensile Testing Per ASTM E8; Brinell Hardness Per ASTM E10; Charpy Testing Per ASTM E23
* AF Global manufactures materials according to a certified Quality Management System conforming to ISO 9001:2015 and PED certified to 2014/68/EU, Annex 1, Section 4.3.
Unless otherwise noted on PO, Dimensions of all standard pipe flanges and flange fittings (NPS 1/2 to NPS 24) per ASME B16.5-2020 , large diameter steel flanges (NPS 26 to 60) per ASME B16.47-2020, Orifice flanges per ASME B16.36-2020.
* AFGlobal uses the latest edition of ASTM E29 (Standard practice for using significant digits in test data to determine conformance with specifications), to claim conformance or nonconformance to a specification. When a accept/reject determination is required AFGlobal uses a test uncertainty ratio (TUR) of 1:1 or higher as acceptable.
* DNV PED Certificate No 10000425664-PA-ACCREDIA-USA Valid thru December 03, 2023.

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Dusty Bannon - Quality Analyst

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